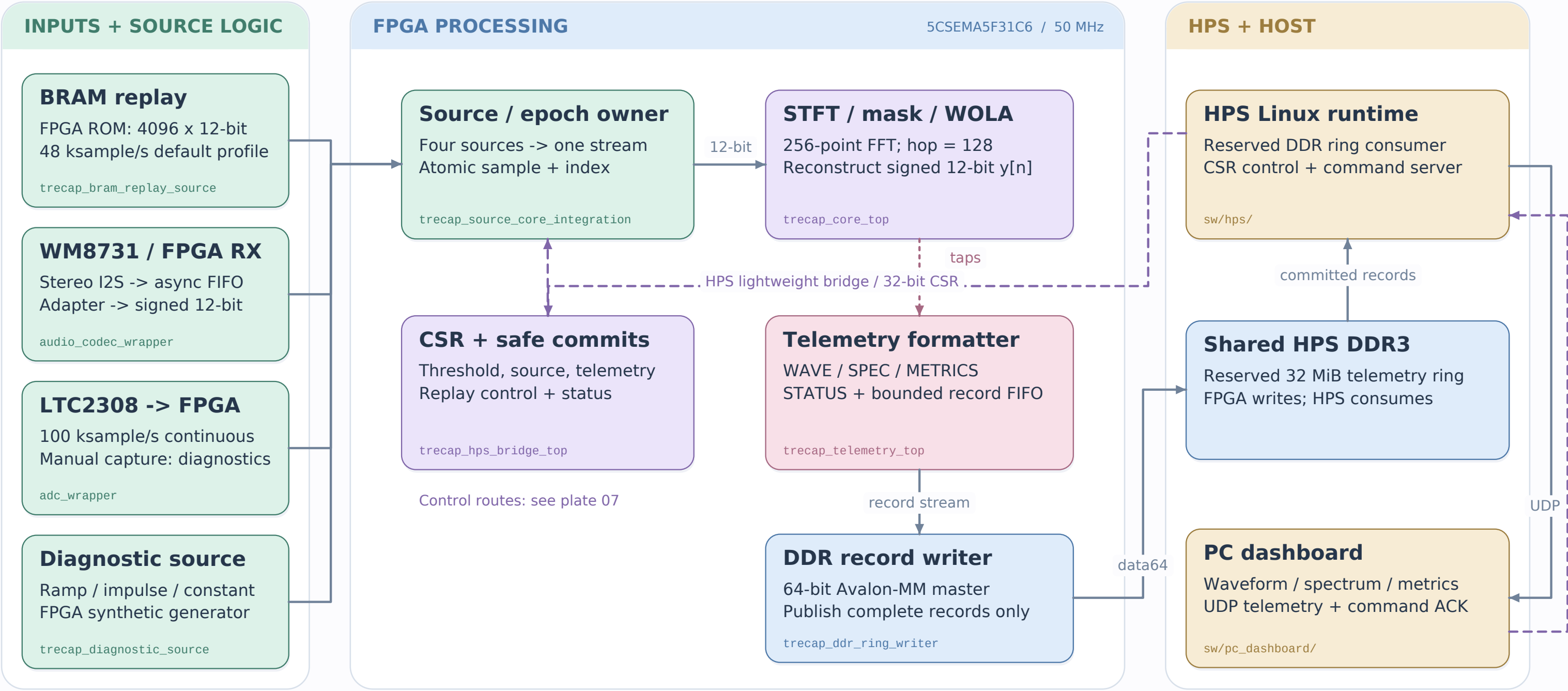


System architecture

A fixed-point FPGA processing path with independent observation, host transport and control.



OFFLINE REFERENCE MODEL

sw/reference_model/ defines the arithmetic reference.
Frozen coefficients populate FPGA ROMs.

CORE OWNERSHIP

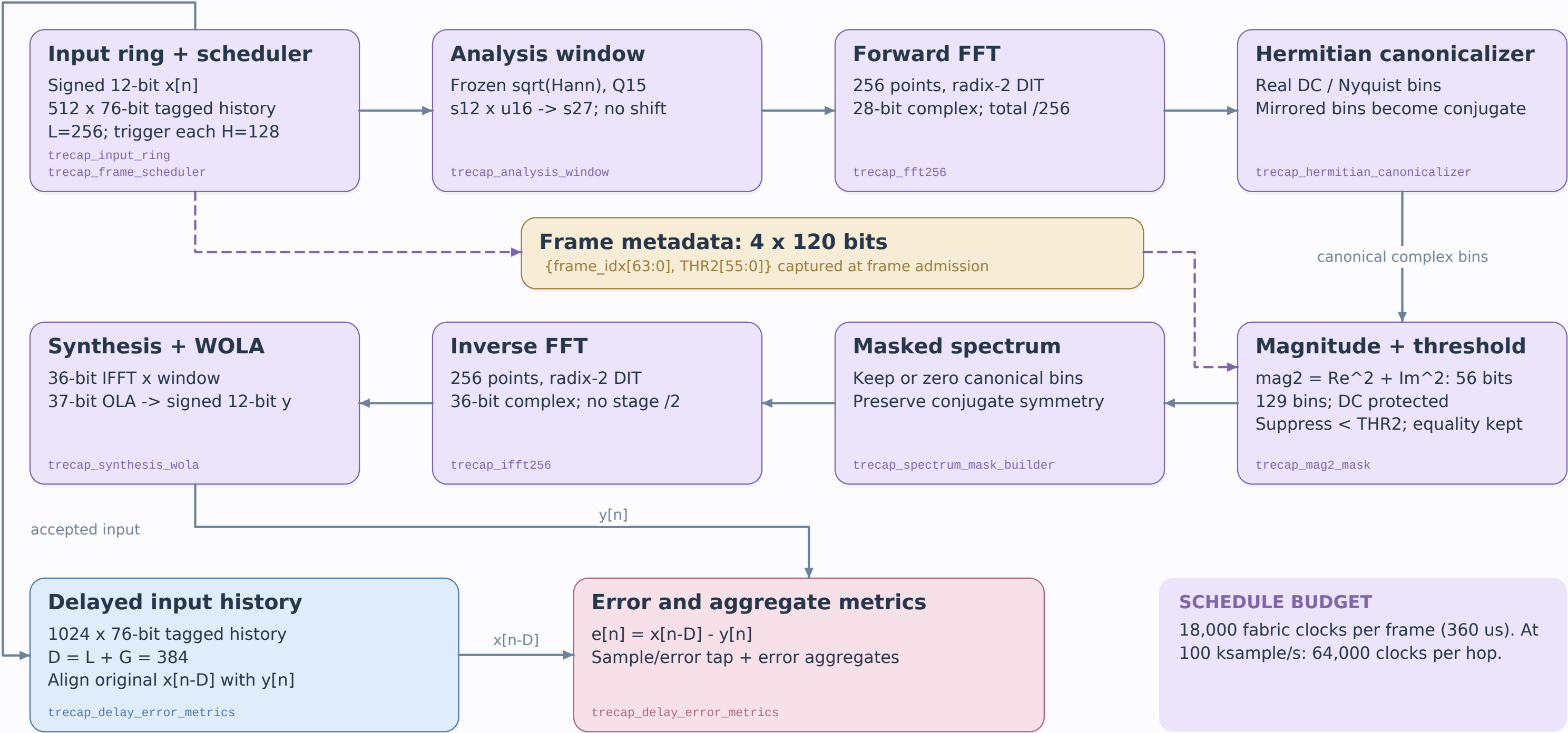
FFT, mask, IFFT and reconstruction execute in FPGA.
Telemetry congestion does not drive core ready.

DEFAULT BRING-UP PROFILE

BRAM zero vector, STATUS-only telemetry,
manual replay start; LINE-OUT disabled.

DSP datapath and frame ownership

The numerical path preserves the fixed 256 / 128 / 384 geometry and frame-owned threshold snapshots.



ARITHMETIC CONTRACT

Round to nearest, ties away from zero; saturation at defined boundaries.

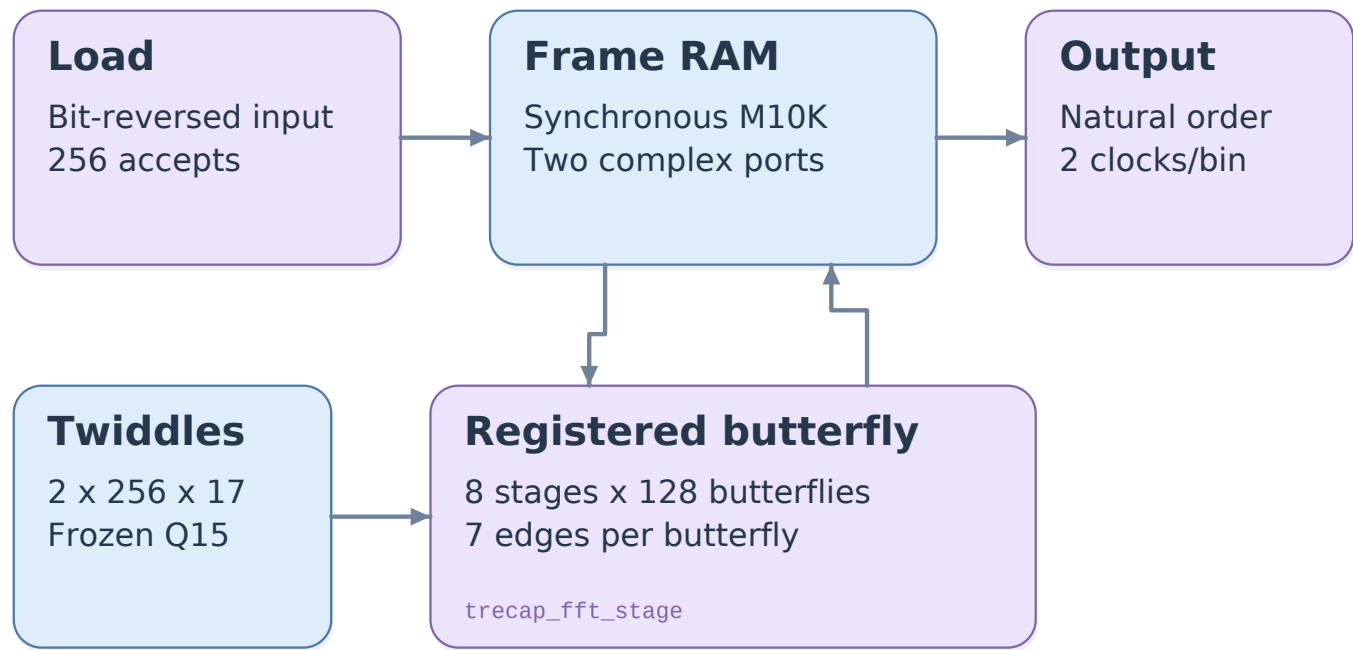
ALIGNMENT

D = 384 samples: 8 ms at 48 ksample/s; 3.84 ms at 100 ksample/s.

Transform engines and storage

Separate iterative FFT and IFFT engines reuse one registered butterfly per engine.

EACH TRANSFORM ENGINE



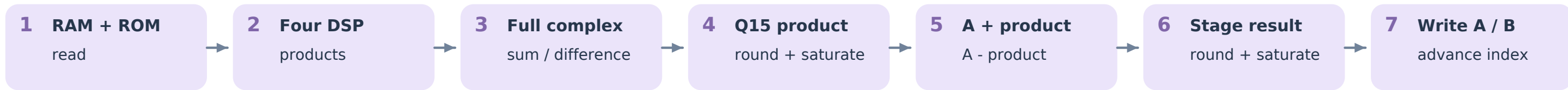
WIDTHS + ENGINE SERVICE

QUANTITY	FFT	IFFT
Work RAM geometry	256 x 56	256 x 72
Data / twiddle component	28 / 17 bits	36 / 17 bits
Four signed products	45 bits each	53 bits each
Full product sum	47 bits	55 bits
Butterfly sum	30 bits	38 bits
Final stage shift	1 bit	0 bits

256 load + 7168 compute + 512 output = 7936 clocks

Local count assumes a complete, available frame and a ready output.

SEVEN-EDGE BUTTERFLY TRANSACTION



SYNCHRONOUS STORE OWNERSHIP

Load, compute and output phases do not overlap inside one engine. Reset invalidates ownership; frame RAM payload is not bulk reset.

OTHER CORE STORES

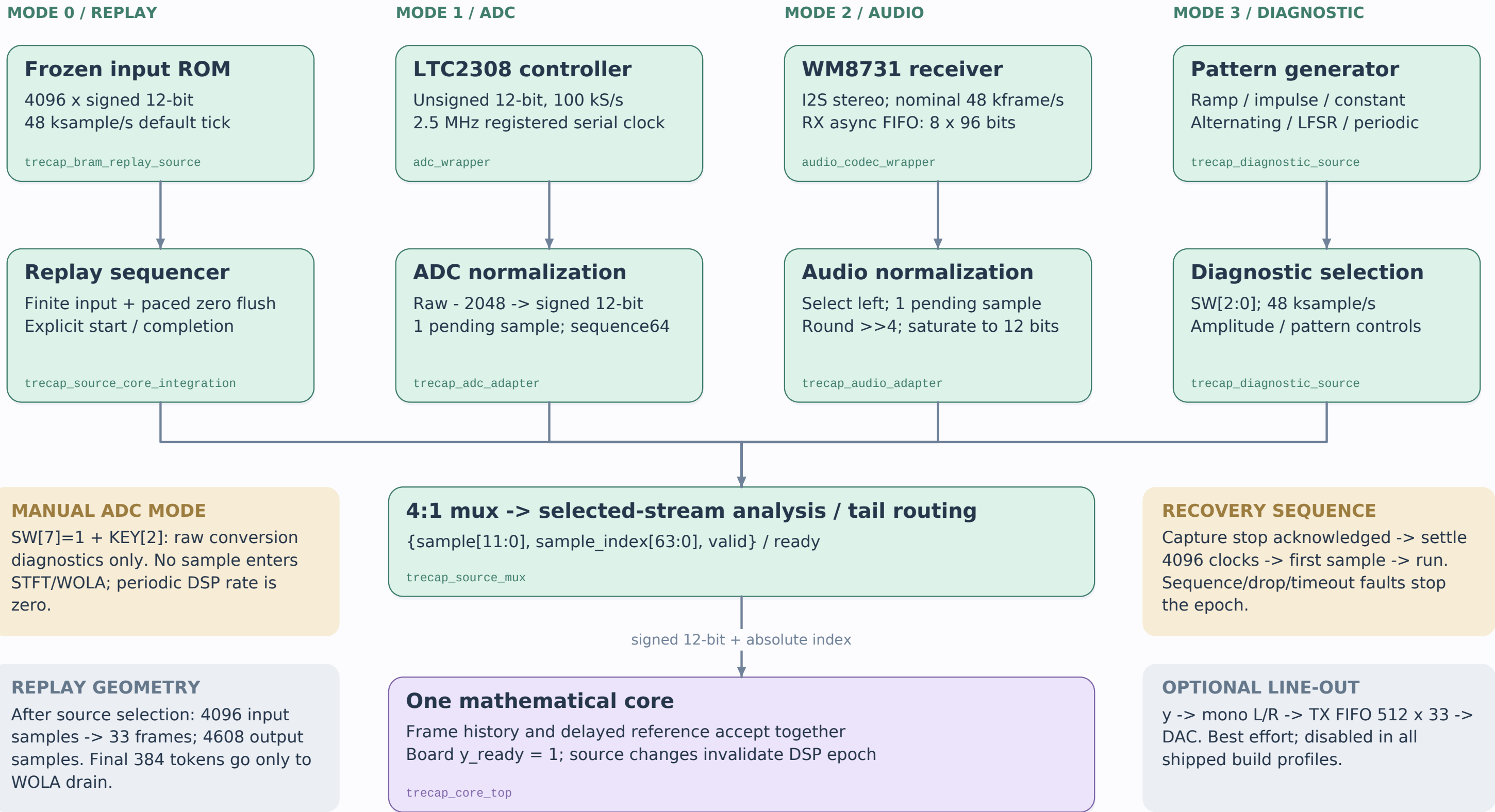
Canonicalizer 256 x 56; input history 512 x 76; delayed history 1024 x 76; WOLA frame 256 x 36 and OLA ring 384 x 37.

FIXED COEFFICIENT PROVENANCE

Analysis window: combinational 256 x 16 ROM. Synthesis: separate synchronous 256 x 16 M10K. Twiddles: 256 x 17 per component; frozen artifacts.

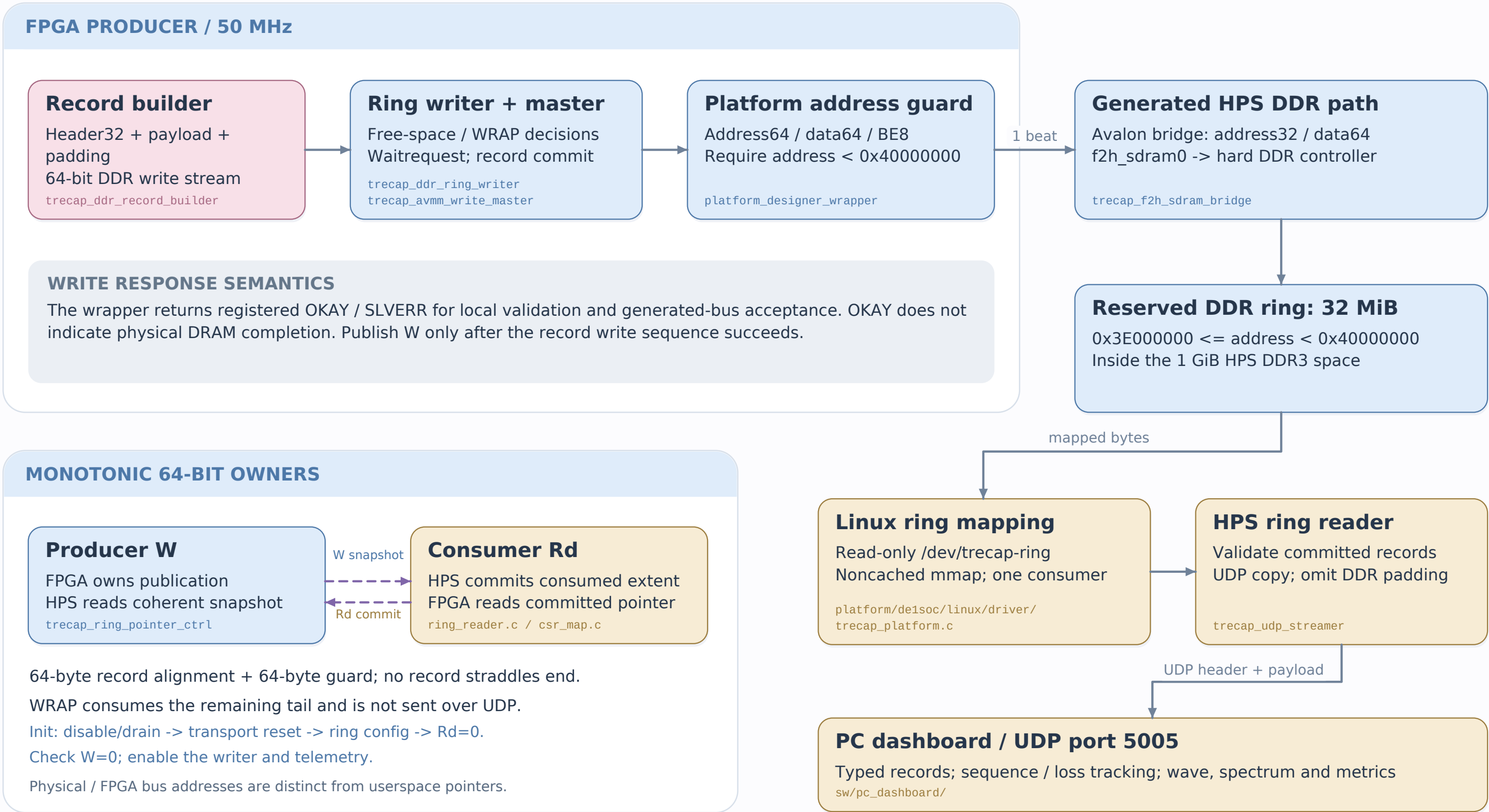
Sample sources and stream epochs

Four physical or synthetic sources share one normalized interface and one DSP history owner.



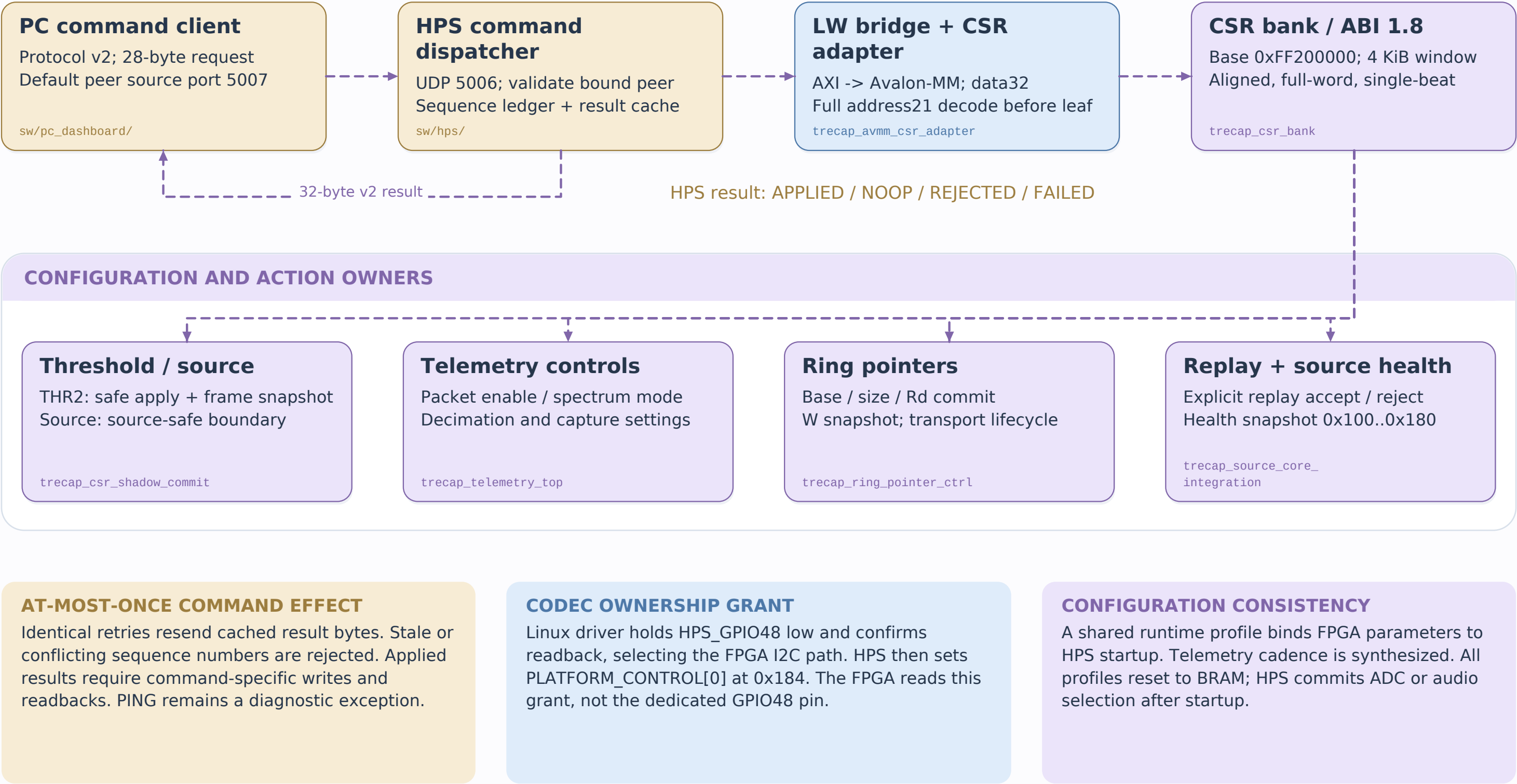
DDR transport and pointer ownership

FPGA produces committed records; Linux consumes them from a reserved, noncached ring.



Commands, CSRs and safe configuration

Control moves from the desktop to HPS software, then across the lightweight bridge into explicit FPGA owners.



Clock, reset and CDC boundaries

The 50 MHz fabric owns numerical processing; external audio serialization crosses through dedicated async FIFOs.

